

Shantanu Sharma

AI/ML Engineer

Gurugram, India | [linkedin.com/in/mathoholic](https://www.linkedin.com/in/mathoholic) | mathoholic.com | github.com/Mathoholic

AI/ML Engineer focused on building production-grade AI systems and scalable backend infrastructure. Experienced in RAG pipelines, LLM-powered search, and recommendation systems, along with event-driven architectures and AWS-based services. Delivered end-to-end products across SaaS, real-time communication, offline navigation, and climate analytics. Strong focus on system design, performance, and deploying AI in real-world applications.

Technical Skills

Languages: Python, Go, SQL, NoSQL
Backend Frameworks: FastAPI, Django, Flask, Gin
Databases: PostgreSQL, MySQL, MongoDB, SQLite, Snowflake
DevOps & Cloud: Docker, AWS (EC2, RDS, S3, ECS), Azure DevOps
Messaging & Queues: RabbitMQ, Celery, WebSockets
Tools & Libraries: NumPy, Pandas, Matplotlib, LangChain
Core Concepts: CI/CD, ETL, Data Warehousing, Microservices, LLM Systems

Experience

AI/ML Engineer – Centocode Technologies

Sep 2024 – Present

- Built RAG-based AI systems using LLMs and ChromaDB, reducing information retrieval time by ~40% through optimized embedding and retrieval pipelines
- Developed natural language search systems for structured datasets, improving accessibility for non-technical users
- Designed and deployed recommendation engines using NLP and content-based filtering, integrated into a React Native app
- Built and maintained vector data pipelines, enabling low-latency embedding generation and retrieval
- Built backend services for a **WhatsApp API SaaS platform**: webhook ingestion, message workflows, and campaign execution; implemented RabbitMQ-based event-driven architecture processing ~150K+ messages/month with retry mechanisms and dead-letter queuing
- Developed an **LLM-based Interview Platform** (AWS: Chime, EventBridge, SQS, RDS): enabled real-time interview sessions with asynchronous evaluation pipelines
- Worked on an **Offline Navigation System** (Ubuntu, C++, QML, OSRM): implemented routing and GPS tracking in fully offline environments
- Contributed to **ClimAnalytix** (Climate Analytics Platform): built pipelines for climate datasets (rainfall, temperature, crop data) and enabled geospatial analytics for risk assessment across regions in India

Senior Data Scientist – 1ViewApps

Feb 2024 – Aug 2024

- Built AI-powered features including search, chatbots, and data visualization using LLMs and LangChain
- Designed modular ETL pipelines for structured and semi-structured data from multiple sources
- Implemented CI/CD pipelines using AWS and Azure DevOps
- Automated data workflows using Airflow, improving reliability of daily data refresh cycles

Associate Data Scientist – KDataScience Solutions Pvt. Ltd.

Jun 2023 – Feb 2024

- Developed data pipelines using Python for large-scale data ingestion and processing
- Built backend services for on-demand data processing
- Designed pipelines for Power BI integration using Python and MSSQL
- Optimized data workflows for performance and scalability

Education

B.A. Mathematics (Hons) – Dr. B.R. Ambedkar University Delhi

2015–2020